CS & IT ENGINEERING

Data types and operators

C Programming

Lecture No.3

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01 Assignment Operator

02 Arithmetic Operator

03 Relational & Logical Operators

printf("______") → as it as

Textual data

printf("10+10"); → 10+10

some int

int a=10;

a

$\boxed{10}$

The value of a is 10

printf("a"); → a

printf("The value of a is %d", a);

① ② ③

scanf

```
int a;

scanf("%d", &a);
```

① "%d" ② &a

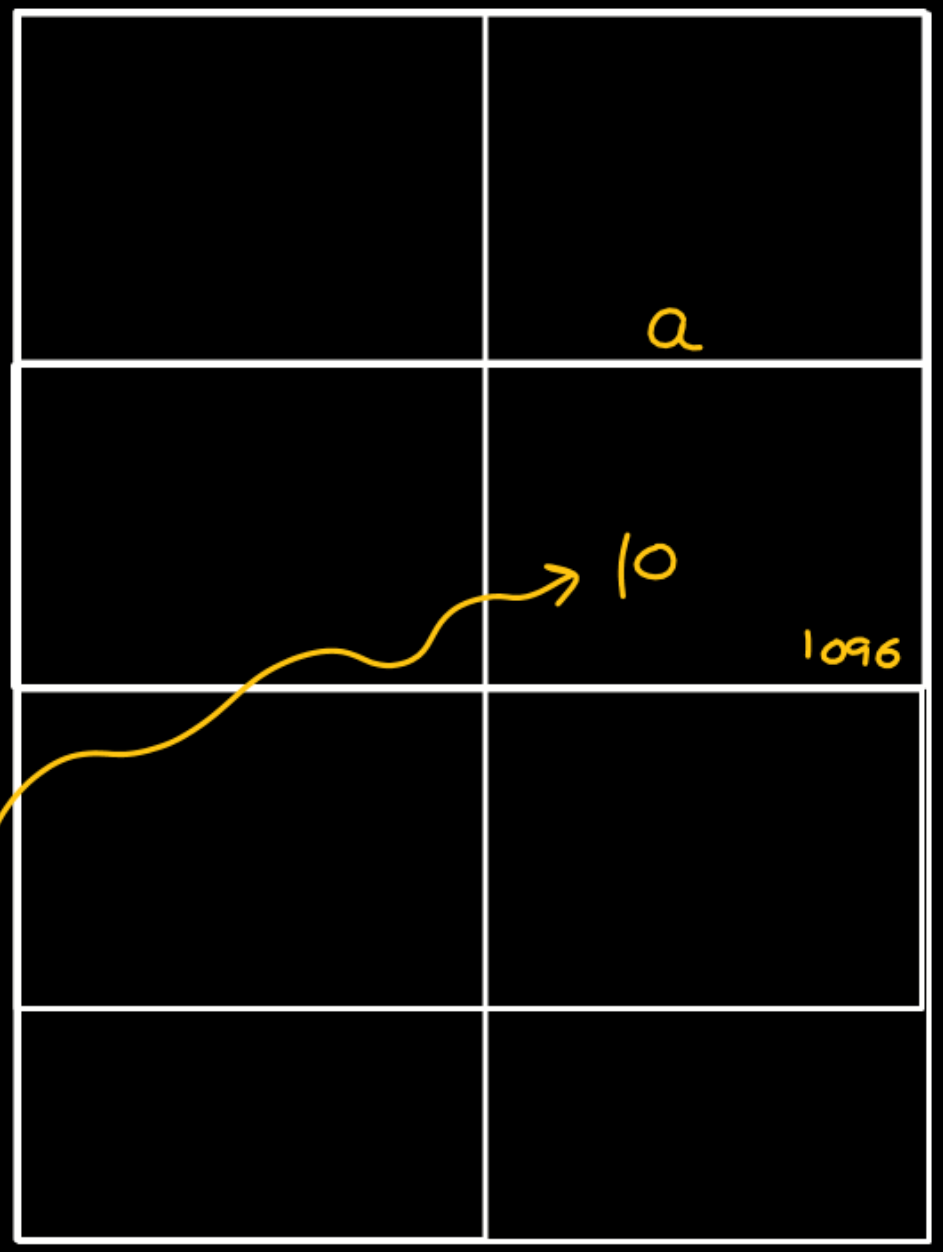
%c → char.

%f → float

%ld → long int

%lld → long long int

10



Operators

Unary Operators

* 1 operand

+12, −7

Binary Operators

* 2 operands

10 + 20

10 x 20

10 / 2

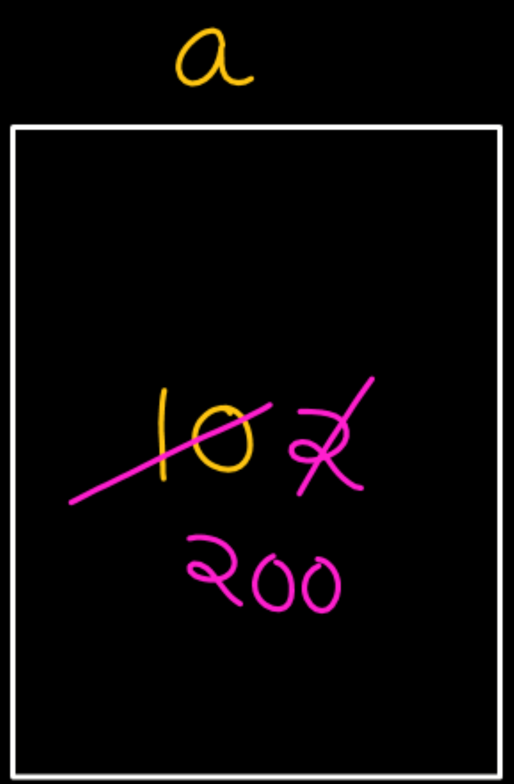
Ternary Operators

* 3 operands

Assignment operator (=)

```
int a=10;
```

$\boxed{a = 2;}$



a=2; ✓

≡

a = 200; ✓

LHS = RHS

```
int a;
2 = a ; ✗
```

```
int a, b=10, c=20;
a = (b+c);
```

a	b	c
30	10	20

```
a = (2+3x7);
```

```
int a=10;
```

Lvalue = Rvalue ;

Lvalue:

- `2=a;` literal ✗ invalid
- `2+3=a;` Expression ✗
- Variable ✓

Rvalue:

- literal ✓ `a = 20;`
- expression `a = 2+3x7;`
- variable `int a, b=10;` `a = b;`

Arithmatic Operators

$\underbrace{\%,\ \times,\ /,}_{\text{binary}}$ $+,\ -$ → unary → binary

printf("%d", $\underbrace{10+20}_{\text{value}}$); valid

a % b : Remainder when a is divided by b.

$12 \% 7 \Rightarrow$ $7\overline{)12}(1$, $12 - 7 = 5$

printf("%d", 12%7); $\boxed{5}$

$2 \% 5 = 2$

$5 \overline{)\,2\,}\,(0 \leftarrow$ Q

$\,0$

$\,2 \leftarrow$ Rem

$12 \% 5 \Rightarrow$

$12.8 \% 5 \Rightarrow$ Error

Both operands of modulas operators must be int type

$-14 \% 5 \Rightarrow$ Standard Not defined

Most of compiler : sign of result $\Rightarrow$ sign of first operator

Result

1. $1234 \% 10 \Rightarrow$

$1230+4$

$$\frac{123 \times 10 + 4}{10} = 4$$

2. $1234 \% 100 =$

$1200+34$

$$\frac{12 \times 100 + 34}{100} = 34$$

last 2 digit

behaviour of operators

$\Rightarrow$ type of operand

$5/2 \Rightarrow 2.5$ ✗

int int

Both are int $\Rightarrow$ Result int

$5/2 = 2$ ✓

float int

$5.0/2 = 2.5$ ✓ float

$5.0/2.0 = 2.5$ ✓ float

$$\underbrace{1/2 \times b \times h}_{0}$$

$\underset{\text{int}}{1}/\underset{\text{int}}{2} = \cancel{0.5} = 0$

$5/2.0 = 2.5$

$$\left.\begin{array}{l} \text{float, int} \\ \text{int, float} \\ \text{float, float} \end{array}\right\} \text{float}$$

$\text{int, int} \Rightarrow \text{int}$

$4/2 \times 2$ → 1 OR 4

① $2 \times 2 \Rightarrow 4$

② $4/4 = 1$

Precedance : ① %, x, /

② +, -

③ =

int a ;

$a = 3 + \underbrace{4 \times 7}_{1}$;

$\times$

$+$

$=$

$a = \underbrace{3 + 28}$

$a = 31$

int a ;

$a = \underbrace{4/2}_{1} \times \underbrace{12}_{2} / \underbrace{6}_{3}$;

same preced.

1 / $\times$

2 =

$\underbrace{4/2}_{1}$

L to R

$a = 2 \underbrace{\times}_{1} 12 \underbrace{/}_{2} 6$

$a = \underbrace{24/6}$

$a = 4$

① %, x, / } L to R

binary ② +, -

③ = R to L

```
int a ;
a = 30.0/2 % 6+4;
printf("%d",a);
```

$a = \underbrace{\overset{\text{float}}{30.0}/\overset{\text{int}}{2}}_{①} \% 6 + 4$

① /, %

② +

③ =

$a = \underbrace{\overset{}{15.0} \% 6}_{②} + 4$

float ② int

Error

operator $\Rightarrow$ value/Result ✓

10 + 20	30
10 / 20	0
10 x 20	200
10 % 20	10

value/result of assignment op.

```
int a = 100;
a = 20;
printf("%d", a = 200);
```

200

200

$\cancel{100}$ $\cancel{20}$ 200

200

int a, b, c;

$a = b = \boxed{c = 4}$;

① what is the value?

$a = \boxed{b = 4}$;

②

$a = 4$

R to L

a	b	c
4	4	4

$10 + 20 + 30$

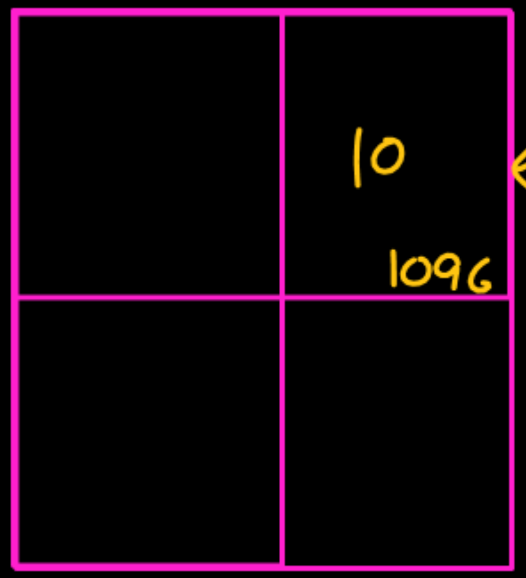
$30 + 30$

07:00 PM

```
void main(){

    int x=10

}
```




```
void main(){
    int a ;
printf("%d",a);
```

Garbage

6:00-9:00pm

$\boxed{F-6}$

```
int a, b, c ;

a = b = 4 = c ;

printf("%d %d %d", a, b, c);
```

Can Lvalue be a literal ✗

Error

Relational Operators

binary

(i) <

(ii) <=

(iii) >

(iv) >=

(v) ==

(vi) !=

$\boxed{10 < 20}$ ⇒ Is 10 less than 20 ?

True

1

```
printf("%d", 10<20);
```

1

20 > 30 : Is 20 greater than 30 ?

False

0

$a <= b$: → Is a less than b ? True

OR

Is a equals to b ?

$10 <= 20$: 1

$20 <= 10$: → F F 0

① printf("%d", 10 <= 5); 0

② printf("%d", 20>10); 1

$a == b$: Is a equals to b ?

$10 == 10$ ✓ 1

$10 == 20$ Is 10 Equals to 20 ? ✗ O/P 0

$a != b$: Is a not equals to b ?

$10 != 20$ Is 10 not equals 20 ? ✓ 1

$10 != 10$ Is 10 not equals 10 ? ✗ 0

The result of every relational operator $\Rightarrow$ 0, 1 Either 0 or 1

```c
int a;

a = 5>4>-1>2>3<0;

printf("%d",a);
```

True

$a = \boxed{5>4} > -1 > 2 > 3 < 0;$

T

$a = \boxed{1 > -1} > 2 > 3 < 0;$

$a = \boxed{1 > 2} > 3 < 0;$

$a = \boxed{0 > 3} < 0;$

$a = \boxed{0 < 0};$

$a = 0$

unary ⇒ +, -

$$\left.\begin{array}{ll} \text{1} & +, - \text{ (unary)} \\ \text{2} & \times, /, \% \\ \text{3} & +, - \\ \text{4} & <, <=, >, >= \\ \text{5} & ==, != \end{array}\right\} \text{L to R}$$

6 = (R to L)

```
int i;
i = 3;
```

```
int i;
i = printf("Hello");
```

5

①

int a; declaration statement ⇒ No. of symbol printed

① Statements

② expression

a = 10; assignment statement

Statement having some value

```
Hello_
```

```c
#include<stdio.h>
void main(){

        12;
        13.78;
        2+3x4;
        }
```

Logical operators

1. Logical AND (&&) } binary
2. Logical OR (||) }
3. Logical NOT (!) } unary

$a \times b = \text{non-zero}$

a	b		
non-zero	non-zero		
0	non-zero	⇒	0
non-zero	0	⇒	0
0	0	⇒	0

a && b:

a	b	a && b	O/P
(non-zero) T	non-zero T	non-zero T	1
F	T	F	0
T	F	F	0
F	F	F	0

```
printf("%d", 12 && 3);            1

printf("%d", -12 && 3.78);        1

printf("%d", 0 && 33);            0

printf("%d", 12.38 && 3.48)       1
```

Logical OR (||)

OR – choice

a||b : if atleast one operand is non-zero(True) : 1
Otherwise ⇒ 0

a	b	a||b
F	F	F
F	T	T
T	F	T
T	T	T

```
printf("%d", 12||1);            1
printf("%d", -12||0);           1
printf("%d", -12.78||3.46);     1
printf("%d", 0||0.0);           0
printf("%d", 0||-12);           1
```

Logical NOT (!)

NOT (True) $\Rightarrow$ False

NOT (False) $\Rightarrow$ True

!5 $\Rightarrow$!(non-zero) $\Rightarrow$!(True) $\Rightarrow$ False $\Rightarrow$ 0

!0 $\Rightarrow$!(zero) $\Rightarrow$!(False) $\Rightarrow$ True $\Rightarrow$ 1

```
printf("%d", !(-12));
```

1. $-, +, !$ (unary)
2. $\times, /, \%$
3. $+, -$
4. $<, <=, >, >=$
5. $==, !=$
6. $\&\&$
7. $||$
8. $=$ R to L

L to R

$\checkmark$

```c
int a;
a = 4==4==1;
printf("%d",a);
```

1

$a = \boxed{4==4} == 1$

$a = \boxed{1 == 1}$

$a = 1$

8 : 30 PM


THANK YOU GW
SOLDIERS !